

PLC Timer Instructions

Pages 8–10 Notes

1. PLC Timers: Introduction

A **PLC timer** is a software instruction that provides time-delay functions similar to mechanical or electronic timing relays. PLC timers perform timing operations inside the control program without any moving parts.

All PLC timers are **output instructions** and are placed at the right side of a ladder logic rung.

Why PLC Timers are Preferred

- Higher accuracy and reliability
- No mechanical wear
- Easy modification through software
- Compact and flexible control logic

2. Types of PLC Timers

2.1 TON: Timer ON Delay

The **TON (Timer ON Delay)** instruction starts timing when the rung condition becomes TRUE. The output turns ON only after the preset time has elapsed.

- Timing starts when rung becomes TRUE
- Accumulated value resets automatically when rung becomes FALSE
- Software equivalent of ON-delay relay

Example: Press START → motor starts after 5 seconds.

2.2 TOF: Timer OFF Delay

The **TOF (Timer OFF Delay)** instruction starts timing when the rung condition changes from TRUE to FALSE. The output remains ON during the delay period.

- Output turns ON immediately when rung is TRUE

- Timing starts when rung goes FALSE
- Software equivalent of OFF-delay relay

Example: Switch OFF → fan continues running for 5 seconds → then stops.

2.3 RTO: Retentive Timer ON

The **RTO (Retentive Timer ON)** instruction accumulates time when the rung is TRUE and **retains the accumulated value** even if the rung goes FALSE or power is lost.

- Accumulated value is not reset automatically
- Requires a RESET (RES) instruction
- Used for total run-time or batch processing

Example: A machine must operate for a total of 30 minutes, even if power interruptions occur.

3. RESET (RES) Instruction

The **RES (Reset)** instruction is used to reset the accumulated value of a retentive timer (RTO) back to zero.

- Clears accumulated value
- Resets done (DN) and timing (TT) bits
- Mandatory for retentive timers

RTO timers never reset automatically. A RES instruction is required.

4. Quantities Associated with PLC Timers

4.1 Time Base

The **time base** is the smallest unit of time used by a PLC timer to count elapsed time.

Common time bases:

- 1 ms (0.001 s)

- 10 ms (0.01 s)
- 100 ms (0.1 s)
- 1 second

4.2 Preset Value (PRE)

The **preset value** specifies the number of time-base intervals required for the timer to time out.

$$\text{Total Timing Interval} = \text{Preset Value} \times \text{Time Base}$$

Preset = 100, Time base = 0.1 s

$$100 \times 0.1 = 10 \text{ seconds}$$

4.3 Accumulated Value (ACC)

The **accumulated value** represents the amount of time that has already elapsed since the timer started.

- Starts from zero
- Increments each time the time base occurs
- Compared continuously with preset value

When **ACC = PRE**, the timer times out and the done bit (DN) becomes TRUE.

5. Representation of Timers in Ladder Logic

5.1 Coil-Formatted Timer Instruction

- Resembles a standard output coil
- Timer parameters are written beside the coil
- Used mainly in early PLC documentation

5.2 Block-Formatted Timer Instruction

- Most common representation in modern PLCs
- Clearly displays timer number, preset, and accumulated value
- Easier to debug and understand

Key Exam Points

- PLC timers are software equivalents of timing relays
- TON provides ON-delay, TOF provides OFF-delay
- RTO retains accumulated time and requires RES
- Total delay = preset value $\times$ time base
- Timers are represented in coil or block format